

June 12, 2023

National Telecommunications and
Information Administration (NTIA)
U.S. Department of Commerce
1401 Constitution Avenue NW
Washington, D.C., 20230

RE: AI Accountability Policy Request for Comment – Docket No. 230407–0093

Credo AI is a Responsible AI Governance Platform that has been a driving force for change within the global AI ecosystem, pushing for greater governance of AI and its responsible development since its inception in 2020. Credo AI's CEO, Navrina Singh, serves as a Member of the National AI Advisory Committee and an OECD AI expert on risk and accountability, and has dedicated her career to advocating for Responsible AI Governance and its safe development and deployment in the United States and around the world. Credo AI is a software tool provider; we have built a Responsible AI Governance platform that supports AI governance in companies of all sizes, across different industries ranging from financial services to insurance, human resources, federal government/procurement, and more. Credo AI is well equipped to answer the NTIA's inquiry and address questions on AI Assurance, and we stand ready and available to support the NTIA and the U.S. Federal Government in this endeavor.

About Credo AI

Credo AI's mission is to maximize the benefit of AI to humanity, using our leading AI governance platform for AI risk management, compliance, safety, ethics, explainability, fairness and economic opportunity. With Credo AI, organizations are able to adopt and scale AI safely and effectively, oversee and mitigate AI risk (including operational, regulatory, and reputational risks), maintain AI ethics standards and compliance (with both internal policies and external regulation and guidance), and future-proof their organization to enable frontier AI adoption. Powered by a Policy Intelligence Engine, organizations can use Credo's Responsible AI

Platform to ensure every AI use case adheres to their business goals, internal ethical principles, and emerging global regulations and standards, like the EU AI Act and NIST AI Risk Management Framework.

Credo AI agrees with NTIA's statement in its Request for Comment¹ that "real accountability can only be achieved when entities are held responsible for their decisions," and that "assessments and audits, governance policies, documentation and reporting, and testing and evaluation can support this process by proving that an AI system is legal, effective, ethical, safe, and otherwise trustworthy."² Credo AI helps companies of all sizes create "AI assurance," and unlock the innovative potential of AI while protecting companies from new and unknown risks. Founded in 2020, Credo AI has been recognized as a Technology Pioneer of 2022 by the World Economic Forum, in Fast Company's "Next Big Thing in Tech 2022" and as a member of the "Intelligent App 40" by Madrona, Goldman Sachs, Microsoft, and Pitchbook.

Executive Summary

Credo AI stands ready and willing to help NTIA develop a productive AI accountability ecosystem. In our response, we have focused on:

- **Governmental and non-governmental actions to support and enforce AI accountability;**
- **Risks and barriers to accountability measures & corresponding recommendations to address these; and,**
- **Recommendations specific to foundation models and large language models.**

Aside from the final section, our response applies to both traditional machine learning (ML) systems and generative and general purpose (genAI) systems. While genAI systems have received significant attention in recent months, Credo AI believes that traditional ML systems will continue to play a significant role in the AI industry and so it is important to establish a strong governance paradigm to apply to all types of AI.

¹ AI Accountability Policy Request for Comment;

<https://www.federalregister.gov/documents/2023/04/13/2023-07776/ai-accountability-policy-request-for-comment>

² Ibid.

Governmental/Non-governmental Actions to Support and Enforce AI Accountability

Credo AI recommends the adoption of a risk-based and rights-based approach to the governance, oversight, and accountability of AI Systems that is context specific. Not all AI is the same, and requirements for AI assurance should increase pursuant to the risk of the AI system. AI assurance is resource intensive for both evaluators and those conducting the assessments; therefore, Credo AI recommends that NTIA optimize AI governance use.

- **For AI assurance, developers and deployers should provide information including (but not limited to):**
 - AI system development at all stages of the AI lifecycle, including:
 - the model architecture;
 - the data used to train the AI system; and,
 - performance of the AI system;
 - How the AI system is tested for and performs against risks identified in a standard risk taxonomy (as explored below);
 - Some form of attestation regarding the protection of fundamental human rights and freedoms (as necessary and proportionate to a democratic society); and,
 - Statement of the system's intended purpose and best practices/instructions for deployers of the system on how to best achieve those purposes.³

- **Taxonomy of Risk:** Credo AI recommends the NTIA continue to work to develop a taxonomy of AI risk to inform the areas that are most important for an AI developer or deployer to consider when assessing its AI system's potential impact. This taxonomy will provide needed and consistent guidance for developers and deployers of AI systems. Credo AI recommends the following risks be considered in such a taxonomy:

³ An example in the spirit of this recommendation would be Open AI's "[Best practices for prompt engineering with OpenAI API](#)," written by Jessica Shieh

- **Abuse & Misuse:** Pertains to the potential for AI systems to be used maliciously or irresponsibly, including for creating deep fakes, automated cyber attacks, or invasive surveillance systems. Abuse specifically denotes the intentional use of AI for harmful purposes.
- **Compliance:** Involves the risk of AI systems violating laws, regulations, and ethical guidelines (including copyright risks). Non-compliance can lead to legal penalties, reputational damage, and loss of user trust.
- **Environmental & Societal Impact:** Concerns the broader changes AI might induce in society, such as labor displacement, mental health impacts, or the implications of manipulative technologies like deepfakes. It also includes the environmental implications of AI, particularly the strain on natural resources and carbon emissions caused by training complex AI models, balanced against the potential for AI to help mitigate environmental issues.
- **Explainability & Transparency:** Refers to the ability to understand and interpret an AI system's decisions and actions, and the openness about the data used, algorithms employed, and decisions made. Lack of these elements can create risks of misuse, misinterpretation, and lack of accountability.
- **Fairness & Bias:** Arises from the potential for AI systems to make decisions that systematically disadvantage certain groups or individuals. Bias can stem from training data, algorithmic design, or deployment practices, leading to unfair outcomes and possible legal ramifications.
- **Long-term & Existential Risk:** Considers the speculative risks posed by future advanced AI systems to human civilization, either through misuse or due to challenges in aligning their objectives with human values & societal goals.
- **Performance & Robustness:** Pertains to the AI's ability to fulfill its intended purpose accurately and its resilience to perturbations, unusual inputs, or adverse situations. Failures of performance are fundamental to the AI system performing its function. Failures of robustness can lead to severe consequences, especially in critical applications.

- **Privacy:** Refers to the risk of AI infringing upon individuals' rights to privacy, through the data they collect, how they process that data, or the conclusions they draw
 - **Security:** Encompasses potential vulnerabilities in AI systems that could compromise their integrity, availability, or confidentiality. Security breaches could result in significant harm, from incorrect decision-making to privacy violations.
- **Assurance Sandboxes for AI:** For commercial systems, Credo AI recommends creating an “assurance sandbox” where commercial entities can use an iterative process for guideline development with limited indemnity to start. This “assurance sandbox” would trial transparency and mitigation requirements (using voluntary guidelines) with non-financial consequences for violations – essentially a “safe harbor” for sincere mitigation efforts. This sandbox would begin with clear, but voluntary guidelines, as part of an effort to create industry best practices and standards for industry to see what works.
- **Dedicated Oversight:** Credo AI recommends that the USG establish dedicated oversight of the procurement, development, and use of AI. In the immediate near term, the USG can achieve this by empowering existing organizations and properly equipping them with the expertise, staffing, funding, and other relevant resourcing needed in order to provide dedicated oversight of AI systems. In the long-term, the USG could also consider the creation of a new independent federal agency or a Cabinet-level position with oversight authority of AI systems. The empowerment of existing agencies or the creation of a new independent federal agency or a Cabinet-level position with the authority to regulate AI should not prohibit the USG from mandating transparency reporting and disclosures from entities throughout the AI value chain here and now (including foundation model providers and downstream developers), so that the fundamental human rights of users can be protected. Lastly, nothing should prevent industry from acting now to do the right thing – industry can be transparent about the AI model architecture, the data used to

train the AI system, its performance, and what its intended and unintended consequences are – even in the absence of dedicated oversight, industry has a responsibility to provide this level of transparency. Dedicated oversight of the AI industry should ensure that AI systems created and employed in the United States are safe, secure, and equitable, and have comprehensive guardrails to identify and curb misuse. Credo AI believes that strong precedents for the establishment of a regulatory body to mitigate these harms and the risk of harm can be found in other high risk sectors (such as pharmaceuticals, healthcare, transportation, and finance), which are all subject to regulatory oversight by sector-specific bodies.

Risks and Barriers to Accountability Measures & Corresponding Recommendations:

Credo AI agrees with NTIA that “there are already existing concerns about the validity of some audits and assessments” that can create barriers to providing adequate and meaningful accountability, for example:

- **Lack of Audit and Governance Workforce Roles:** The lack of a certified workforce who are trained in AI governance, assurance and auditing (and limited corresponding training and resources for staff) could present a significant barrier to providing adequate and meaningful accountability as part of the AI assurance ecosystem at the federal government level. Credo AI encourages NTIA to further fund workforce development for these new roles and positions focused on AI assurance, as well as training and accreditation in the AI assurance and auditing ecosystem.
- **Lack of Metrics and Evaluation Standards:** Credo AI recommends significantly increased investment in federal government resources – especially for entities like National Institute of Standards and Technology (NIST) – to develop evaluation standards for AI systems, especially large language models.

- Narrow Scope:** if auditing and assessment criteria are scoped too narrowly, there is a risk of creating a “false sense” of assurance. For instance, given the pressure that has been placed on legislatures to avoid “hindering innovation,” regulations often limit which companies are actually covered based on the degree of autonomy of the system. Credo AI encourages NTIA to avoid overweighting degree-of-autonomy as a criteria for determining whether a system should be covered by regulation. There are a large number of non-“autonomous”, non-“automated” systems that, due to their potential risks, should be subject to human review and extensive AI assurance checks provided by outside audits and assessments. In the generative AI space, specifically, as of this writing, there are a small handful of foundation model “providers.” Mandating transparency reporting for “providers” only would exclude a growing number of companies that would be considered “downstream developers” – organizations which have fine-tuned the LLM to suit their purpose – whose product or products pose risks similar to those posed by foundation models and thus should be required to provide transparency reporting, undergo audits, and provide assessments in line with those required of providers.
- Goodhart’s Law:** Goodhart’s law can also be a risk for AI assurance and assessments. “Goodhart’s Law⁴” is characterized as an understanding that “Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes⁵,” or more simply, “When a measure becomes a target, it ceases to be a good measure.” For this reason, Credo AI recommends that NTIA consider a set of metrics and statistical measures to assess AI systems – not just one – to avoid a “race to the bottom” by companies seeking to simply “lower their overall score” to achieve a percentage or particular metric for evaluation of adherence to Responsible AI principles.

⁴ Named after British economist Charles Goodhart, who is credited with expressing the core idea of the adage in a 1975 article on monetary policy in the United Kingdom.

⁵ “Goodhart’s Law,” Wikipedia The Free Encyclopedia, https://en.wikipedia.org/wiki/Goodhart%27s_law

- **Access to Data:** As NTIA has rightly noted, policies adjacent to “AI system specific” legislation and regulation – namely policies in the data and consumer protection space – also have an effect on the ability of auditors to conduct assessments, and the validity of those assessments. There are two issues related to access to data that arise in the context of audits of AI systems: (1) how much access to data the auditor is granted; and, (2) what details become publicly available as a function of the audit. Higher risk AI systems typically necessitate a greater level of access by the auditor, and AI systems that pose a higher risk to the public or are likely to impact a broader segment of society should require a higher degree of public transparency. For example, in the SOC2 auditing standard, auditors must be provided with a significant level of access to organization systems to ensure compliance. Organizations, however, do not need to publish audit results in order to maintain compliance; they retain the choice of third parties with which to share audit reports. Similarly, in the financial sector, auditors are typically granted a significant degree of access to the accounting information of the organizations they are auditing. In contrast, certain auditing results – such as revenue and expenditure data in publicly traded corporations – are published by the audited organization in accordance with legal mandates.

Recommendations Specific to Foundation Models and Large Language Models:

In light of the recent focus on generative AI systems, Credo AI urges NTIA to consider and support the following recommendations to support increased transparency and oversight of Foundation Models (FMs) and Large Language Models (LLMs), namely:

- **Model-Level Transparency Disclosures by FM/LLM Providers**
 - Transparency disclosures that should be made available to downstream application developers and to the appropriate regulatory or enforcement body within the United States Government (USG) – not the general public – to ensure they are fit for purpose.

- Organizational-Level Process (and Use) Disclosures by FM/LLM Providers with Corresponding Enforcement
 - Organizations should be required to disclose (to the general public and corresponding enforcement agency) if and how they are using LLMs/FMs. Beyond transparency disclosures at the model level, LLM/FM providers must institute organization-level policies for responsible use and development, including investment in AI safety and governance, and processes to ensure full lifecycle governance.
- Non-disclosure related recommendations:
 - Create a new lab at NIST dedicated to AI safety and governance research (distinct from AI capabilities research and development).
 - Public commitment by FM/LLM providers to invest 10% of AI R&D investments in AI safety and governance (people, processes and tools).
 - Credo AI recommends that “AI Safety & governance research,” (distinct from AI capabilities research and development) be considered in addition to the new investments in “climate, agriculture, energy, public health, education, and cybersecurity” announced by the White House in May 2023 as part of the \$140 million in National Science Foundation funding to launch seven new National AI Research Institutes.⁶
 - In addition to general safety and governance research, dedicated research efforts should include work on explainable AI methods, including neural network-specific methods like “mechanistic interpretability.” The goal of this research should be answering the question, “Why do neural networks do what they do?”

⁶ FACT SHEET: Biden-Harris Administration Announces New Actions to Promote Responsible AI Innovation that Protects Americans’ Rights and Safety, 4 May 2023, <https://www.whitehouse.gov/briefing-room/statements-releases/2023/05/04/fact-sheet-biden-harris-administration-announces-new-actions-to-promote-responsible-ai-innovation-that-protects-americans-rights-and-safety/>

Credo AI appreciates this opportunity to comment on these issues and NTIA's efforts toward an effective AI assurance ecosystem. We welcome any further opportunity to provide resources or information to assist in this important effort. If you have any follow-up questions regarding these comments and recommendations, please contact Credo AI's Director of Global Policy, Evi Fuelle (evi@credo.ai).

Sincerely,

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